Alvaro Pastor

Professor, Master's Degree in Videogame Design and Development
Department of Computer Science, Multimedia, and Telecommunication
Open University of Catalonia, Spain
alvaropastor@uoc.edu | +34 613173089 | ORCID 0000-0002-0467-8260

1. EDUCATION

2026 Ph.D., Network and Information Technologies
Open University of Catalonia, Spain.
Dissertation: "Virtual and Augmented Reality Investigations of Episodic Memory: The Role of Locomotion, Spatial Features and Emotion".

2015 M.Sc., Cognitive Systems and Interactive Media
Pompeu Fabra University, Spain.

2013 Architect
Applied Sciences University, Peru.

2. PROFESSIONAL APPOINTMENTS

2020–Present. Professor, Master's Degree in Videogame Design & Development
Open University of Catalonia, Spain.

2023–2024. Technical specialist, Virtual Reality Design & Development
EventLAB, Department of Clinical Psychology and Psychobiology, Universitat de Barcelona, Spain.

2022–2024. Professor, Degree in Videogame Design & Development
Centre for Image and Multimedia Technology, Universitat Politècnica de Catalunya, Spain.

3. GRANTS & FUNDED RESEARCH

2024–2027 Inhabiting the Hybrid: **Contributions from Artistic Research in Interactive and Immersive Media** (PID2021-128875NA-I00). MCIN/AEI + ERDF. *Role: Researcher. [Active]*

2024–2025 NeuroScent VR. Research Accelerator Grant RA405. Open University of Catalonia. *Role: Lead Developer. [Active].*

2023–2024 EnvironMENTAL (101057429). Horizon Europe / Universitat de Barcelona. *Role: Lead VR Designer & Developer. [Completed]*

2021–2022 The Memory Palace Project. Innovative Information and Communication Technologies Grant. Barcelona Culture Institute. *Role: Principal Investigator. [Completed].*

2019 Industrial Doctorates Grant. Generalitat de Catalunya. *Role: PhD Fellow. [Awarded, not executed].*

4. SCIENTIFIC CONTRIBUTIONS

Selected Peer-Reviewed Publications:

Pastor, A.; Bourdin-Kreitz, P. (2026). Designing memorable XR under stress: An empirically grounded criterion for long-term memory binding. In CHI 2026 Workshop on XR for Challenging Environments (XR4CE): Enabling Human Performance and Agency under Stress. ACM.

Pastor, A.; Bourdin-Kreitz, P. (2026). Comparing Episodic Memory Binding Outcomes After Emotion Induction in Virtual Reality. *Virtual Reality*. https://doi.org/10.1007/s10055-026-01364-9

Pastor, A.; Bourdin-Kreitz, P. (2026, *in review*). Synthetic generation of photorealistic 3D characters for face-based episodic memory measures. *ACM Trans. Appl. Percept.* https://doi.org/10.31219/osf.io/4jf2g_v1

Pastor, A. (2025). Augmenting Reality: On the Shared History of Perceptual Illusion and Video Projection Mapping. Estudis Escènics - Quaderns del Institut del Teatre, 50, 326 - 343. e-ISSN: 2385-362X.

Pastor, A.; Bourdin-Kreitz, P. (2024). Comparing episodic memory outcomes from walking augmented reality and stationary virtual reality encoding experiences. *Scientific Reports, 14*, 1-23. https://doi.org/10.1038/s41598-024-57668-w

5. TECHNICAL & METHODOLOGICAL EXPERTISE

Immersive & interactive Systems: AR/VR design and development. Interactive systems engineering. Full-stack interactive web technologies. User experience optimisation for immersive environments.

Cognitive science methods: Creation of ecologically valid measurement frameworks for immersive environments. Development and validation of novel cognitive assessments. Integration of standard psychometrics into experimental designs.

Machine learning & computer vision: Neural network-based 2D/3D synthesis. Computer vision applications. Algorithmic generation of auditory, semantic and visual stimuli.

Hardware integration: Psychophysiological sensor integration in immersive environments. Electromechanical systems engineering for experimental platforms. Multimodal data synchronisation.

Data science & computational methods: Statistical analysis and computational modeling. Behavioural and physiological data processing. Scientific data visualisation.